JI-YOUNG **YANG**

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EDUCATION

PhD in Analytical Chemistry
University Ulm 11/2020 - 11/2025 Ulm, Germany
Thesis : "Mechanistic Insights into Protein-Cosolute Interactions: From Denaturants to Stabilizers in Biopharmaceutical Formulations"
Dissertation submitted (11/2025); currently under review, defense expected in Q1 2026

MSc in Bioinformatics/Molecular Biology (Note 2,1)
University of Applied Sciences Mittweida 2018 - 2020 Mittweida, DE
Thesis: "Bioinformatics approaches for characterization of quality attributes of biopharmaceuticals using statistical and machine learning based methods" (Note 1,0 1,5)

BEng in Bioengineering (Note 1,7)
Incheon National University 2012 - 2017 Incheon, South Korea

EXPERIENCE

PhD Student
Boehringer Ingelheim Pharma GmbH Co. KG 11/2020 – 07/2024 Biberach, Germany

Industrial Ph.D. research in Analytics Development Biologicals, Early Stage Analytics Laboratory

- Developed ML models for spectral deconvolution and pattern recognition in antibodies and excipients characterizations.
- Developed Automated liquid handling and data analysis process with python for protein characterization studies. Data management and lightweight data engineering for daily experimental outputs, supporting structured storage, automation, and reproducible analysis.
- Studies of antibody interactions with and cosolutes using classical MD simulations and analytical chemistry methods.
- Designed protocols for antibody stability studies, including forced degradation and enzymatic digestion.

Master Thesis & Internship
Boehringer Ingelheim Pharma GmbH Co. KG 03/2020 - 10/2020 Biberach, Germany

Thesis: "Bioinformatics approaches for characterization of quality attributes of biopharmaceuticals using statistical and machine learning based methods" (Note 1,0)

- Developed ML model for pattern recognition of surfactant degradation profiles in drug formulations, enabling classification and clustering of degraded lipid chains from UPLC-CAD data
- Developed neural network model to predict solvation energy of electrolytes based on their atomistic properties achieved by conceptual Density Functional Theory principles [1,2]

Working Student in Quality Control
BIOTYPE GmbH 01/2019 – 10/2019 Dresden, Germany

- Data validation and Testing Quality Control of molecular diagnostic solutions

Trading Specialist for Pure Chemicals
Samchun Pure Chemical Co. Ltd 03/2018 – 09/2018 Seoul, South Korea

- Utilized chemical product metadata, safety documentation (SDS, CoA), and regulatory data (K-REACH, REACH, TSCA) to support compliant import/export of specialty chemicals across Europe, the USA, Japan, and India.
- Analyzed logistics and pricing data to coordinate international shipments, negotiate freight costs, and adapt pricing strategies to changing market conditions.

ACTIVITIES

Exchange program in Global Data Science

Boehringer Ingelheim

01/2024 - 02/2024 Ingelheim, Germany

Internal exchange program in Global Data Science team Health Care Affair & Patient Engagement for project **"AI and Data Science for Global Patent Search & Analytics"**

Co-Lead National Programmed

Nucleate Germany

03/2023 - 01/2024

- As an early member, contributed to its development of Germany into a Core Chapter of the global nonprofit organization *Nucleate*, supporting future biotech entrepreneurs by recruiting experts in the biotech industry, intellectual property, and business development for the "Activator" mentoring program

Exchange program in Biochemistry (German)

University Hohenheim

2016 - 2017 Stuttgart, Germany

Exchange program in Business Management (English)

HfWU Nürtingen-Geislingen

2016 - 2017 Nürtingen, Germany

SKILLS

Languages

- **Korean** (native)
- **English** (fluent C1-C2. Academic and business proficiency with experience in international scientific presentations and publications)
- **German** (fluent C1. Completed Master's degree in German; Professional working proficiency; extensive experience in German-speaking corporate environments)

Programming

- Python, R, SQL : Focused on **Bioinformatics**, **Chemometrics** (RDKit, MDAnalysis), **Machine Learning** (Pytorch, scikit-learn, Tensorflow, JAX) and **Data Science** (seaborn, plotly, pandas, streamlit)
- Data management and engineering
- HPC (High Performance Computing) : Bash and Slurm batch system on Linux HPC clusters
- Cloud Computing : Streamlit, Databricks, Bitbucket

Computatinal Chemistry

Protein-ligand interactions studies, small drug molecules (peptides) conformational changes studies in chemical sequence variants utilizing classical MD simulation and docking tools

- GROMACS, NAMD, OpenMM

Lab techniques

Protein and Formulation Excipients Characterization:

- Spectroscopy: UV/Vis, FTIR, CD and Fluorescence spectroscopy for stability testing and structural analysis.
- Calorimetry and Light Scattering: Differential Scanning Calorimetry (DSC), nanoDSF, and Dynamic Light Scattering (DLS) for evaluating thermal stability and aggregation.
- Chromatography: LC-MS, UPLC-CAD

Software & Tools

MS Office, Excel Visual Basic, SIMCA, Spotfire for industrial analytics

CONFRERENCES & AWARDS

Poster Presentation
Discover BMB
03/2023 Seattle, USA
Abstract 203 ***ASBMB*** (Americal Society for Biochemistry and Molecular Biology)

Oral Presentation
DPG Frühjahrstagung
03/2023 Dresden,Germany
Biophysics Chapter 17 Protein Structure and Dynamics *17.8 Talk*
DPG (Deutsche Physikalische Gesellschaft) Spring Conference

Engagement Individual Award
BI OPEN Global Data Science Challenge
06/2021
Boehringer Ingelheim OPEN global data science challenge for Lung Acoustic Signal Detection

LINKS

Homepage
https://jiyoungyang1.github.io/portfolio/
Linkedin
https://www.linkedin.com/in/ji-young-y-28a828142/

REFEREE

Prof. Dr. Boris Mizaikoff
Universität Ulm
Institute Director and Chair of Institute Analytical Chemistry and Biochemistry
boris.mizaikoff@uni-ulm.de
PD. Dr. Jens Smiatek
Boehringer Ingelheim GmbH & Co. KG
Digital Lead, Development NCE,
jens.smiatek@boehringer-ingelheim.com
Dr. Oliver Burkert
Boehringer Ingelheim GmbH & Co. KG
Lab Head of Early Stage, Analytical Development Biologicals
oliver.burkert@boehringer-ingelheim.com

PUBLICATIONS

Published 6 peer-reviewed articles (5 first-author) in high-impact physical chemistry journals (Q1-Q2).

Peer-reviewed published full articles

1. Thermal Stability of IgG4 Monoclonal Antibodies: Ambiguous Roles of Co-Solutes, Excipients and Salts in Conformational and Colloidal States Ji Young **Yang**, Oliver Burkert, Boris Mizaikoff, Jens Smiatek Mol. Pharmaceutics 2026, Article ASAP
2. Influence of Co-Solutes and Solvents on Diffusion Interaction Parameters in Multicomponent Solutions: New Insights through the Kirkwood–Buff Theory Jiyoung **Yang**, Matthias Brosz, Niklas Adebar, Oliver Burkert, Moritz Schulze, and Jens Smiatek* J. Phys. Chem. B 2025, 129, 10381–10391
3. Multidomain Protein−Urea Interactions: Differences in Binding Behavior Lead to Different Destabilization Tendencies for Monoclonal Antibodies. Ji Young **Yang**, Oliver Burkert, Boris Mizaikoff, Jens Smiatek, J. Phys. Chem. B 2024, 128, 42, 10408–10416
4. Impact of Urea on Monoclonal Antibodies: Multiple Destabilization and Aggregation Effects for Therapeutic Immunoglobulin G Proteins Ji Young **Yang**, Oliver Burkert, Boris Mizaikoff, Jens Smiatek, ACS Omega 2024, 9, 5, 5517–5522
5. Combination of Explainable Machine Learning and Conceptual Density Functional Theory: Applications for the Study of Key Solvation Mechanisms I-Ting Ho, Milena Matysik, Liliana Montano Herrera, Jiyoung **Yang**, Ralph Joachim Guderlei, Michael Laussegger, Bernhard Schrantz, Regine Hammer, Ramón Alain Miranda-Quintana and Jens Smiatek, Phys. Chem. Chem. Phys., 2022,24, 28314-28324
6. Artificial neural networks for the prediction of solvation energies based on experimental and computational data. Jiyoung **Yang**, Matthias J. Knape, Oliver Burkert, Virginia Mazzini, Alexander Jung, Vincent S. J. Craig, Ramón Alain Miranda-Quintana, Erich Bluhmki and Jens Smiatek. Phys. Chem. Chem. Phys., 2020,22, 24359-24364